Explainable and Multimodal Classification of Breast Cancer Histopathology Images

1st Jonathan Brown
Department of Biomedical Engineering
The Johns Hopkins University
Baltimore, U.S.A.
jbrow365@jh.edu

2nd Zhuoyao (Martin) Xin
Department of Biomedical Engineering
The Johns Hopkins University
Baltimore, U.S.A.
zxin7@jh.edu

3rd Sakshi Singhal
Department of Biomedical Engineering
The Johns Hopkins University
Baltimore, U.S.A.
ssingha9@jh.edu

4th Nnemdi Amanambu
Department of Biomedical Engineering
The Johns Hopkins University
Baltimore, U.S.A.
namanam1@jh.edu

***Abstract*—Breast cancer histopathology classification remains challenging due to subtype imbalance, heterogeneous magnification levels, and limited interpretability of deep learning systems. We present a magnification-aware multimodal framework for hierarchical breast cancer classification on the BreaKHis dataset. The proposed ResCNN achieved a patient-level AUC of 0.925 for benign-v-malignant classification, while Vision Transformer and Swin Transformer architectures demonstrated competitive binary classification performance but limited subtype generalization. Model interpretability was evaluated through comparison of Gradient-weighted Class Activation Mapping (GradCAM) and occlusion analysis, revealing invariance from magnification and broad model attention. These findings support the feasibility of magnification-aware models while revealing limitations of this information's utility and explainability through traditional frameworks.**

***Index Terms*—ML, CNN, Classification, Saliency**

I. Introduction

Cancer, characterized by uncontrolled cellular growth, remains a premier global health crisis, with breast cancer specifically accounting for 2.3 million new cases and 670,000 deaths in 2022 alone [1]. Projections suggest that by 2050, annual cases will rise to 3.2 million, disproportionately impacting regions with limited diagnostic infrastructure [2]. This trajectory underscores an urgent need for scalable, accurate diagnostic tools to mitigate the strain on healthcare systems and reduce preventable mortality.

The clinical gold standard for diagnosis is the histopathological examination of biopsy tissue [3]. Pathologists evaluate features such as cellular morphology, biomarker staining, and mitotic counts to differentiate between benign and malignant states [4]. Despite its "gold standard" status, the process is susceptible to human error; a study of 988 biopsies found a 2.2% false-negative rate, with 64% of those errors attributed to misinterpretation [5]. Diagnostic discrepancies often arise when malignant features, such as atypical hyperplasia or ductal carcinoma in situ (DCIS), mimic benign patterns [6], creating significant uncertainty for clinicians and risks for patients.

Deep learning (DL) offers a potential solution for standardizing these interpretations, yet clinical translation is often hindered by class imbalance, magnification variability, and the "black box" nature of model outputs [7] [8]. These barriers are particularly evident in the BreaKHis dataset, where 7,909 images across four magnification levels (40× to 400×) introduce substantial variations in texture and scale.

Our project addresses these gaps by implementing a Swin Transformer framework designed for binary malignancy classification. By focusing on a model that remains robust across varying magnification levels and incorporating gradient-based visualization (GradCAM), we provide an explainable diagnostic aid. This approach seeks to enhance the reliability of automated screening, supporting pathologists in making rapid, verifiable diagnostic decisions while improving outcomes for patients through more consistent classification.

II. Background

Spanhol et al. demonstrated that patch-based CNNs trained directly on biopsy images can outperform handcrafted pipelines while eliminating brittle feature-selection steps. [9] A CNN computes spatial features via learned kernels,

$$(\mathbf{f} * \mathbf{x})_{i,j} = \sum_{u,v} w_{u,v}\, x_{i+u,\, j+v} \tag{1}$$

followed by nonlinear activation functions and progressively coarsens representation using pooling. These operations allow CNNs to learn diagnosis relevant features (e.g. glandular boundaries) directly from data. Later work extended this paradigm through transfer learning, repurposing pretrained feature extractors, providing competitive accuracy. [10]

More recent architectures such as ResNet and Xception have furthered improvements by decoupling spatial and channel correlation patterns, achieving high performance (F1 $\approx$ 0.90). [11] These findings motivate our choice to employ a ResNet-style encoder that balances depth with computational tractability in addition to Zhang et al's proposed two-stage ensemble classifier in which "easy" samples are handled by an initial model while "ambiguous" cases are deferred for refinement or human review. [12] Zhang's cascade demonstrated that structuring classification hierarchically can substantially improve reliability (97.65%).

Modern DL-based diagnostic systems also depend on a clear mathematical formulation of classification objectives. Most CNN-based pipelines optimize cross-entropy,

$$\mathcal{L}_{\text{cls}} = -\sum_{c} y_c \log p_c, \tag{2}$$

with class probabilities computed via a softmax function,

$$p_c = \frac{\exp(z_c)}{\sum_k \exp(z_k)}. \tag{3}$$

In our setting, the subtype loss is restricted to semantically valid label groups. This is achieved by weighting benign and malignant subtype losses,

$$\mathcal{L}_{\text{sub}} = \frac{n_b \cdot \mathcal{L}_b + n_m \cdot \mathcal{L}_m}{n_b + n_m}, \tag{4}$$

and coupling them to the main diagnostic objective,

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{sub}}, \tag{5}$$

ensuring that subtype gradients propagate only within the correct diagnostic branch.

However, because clinical workflows require interpretable predictions, whitebox techniques are essential. GradCAM identifies the strongest contributing spatial regions to a model's decision using gradient-weighted activations, producing maps that help verify that a model attends to diagnostically relevant structures and reveal failure modes. [13]

III. Methods

A. Pipeline

We developed a magnification-aware deep learning framework for hierarchical breast cancer classification using histopathology images from the BreakHis dataset. The overall pipeline comprises data preprocessing, multimodal representation learning, hierarchical classification, patient-level aggregation, model interpretability, and benchmarking, as illustrated in Fig.1.

The BreakHis dataset includes breast tumor histopathology images acquired at four magnifications (40×, 100×, 200×, and 400×). All images underwent standardized preprocessing, including normalization (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) and stain-preserving data augmentation (random flipping, rotation, cropping, and color perturbation) to enhance generalizability. To mitigate class imbalance across benign and malignant categories and their subtypes, balanced sampling strategies were applied during training. Model evaluation was performed using strict patient-level five-fold cross-validation, ensuring that images from the same patient were never shared across training and evaluation sets.

To leverage multi-scale morphological information, we constructed magnification-aware multimodal embeddings. Each image was encoded using a shared ResCNN backbone followed by an MLP projection head. The MLP head consisted of two fully connected layers with ReLU activation and dropout regularization. Magnification metadata was incorporated via learnable one-hot token embeddings and additively fused with image features to form a unified latent representation. Feature alignment across magnifications was implicitly learned, enabling the model to integrate complementary structural patterns observed at different resolutions while maintaining a shared encoder across all magnifications.

We formulated the classification task as a hierarchical problem consisting of a binary (benign vs malignant) branch and a subtype classification branch. The model jointly optimizes both tasks using a gated mechanism, where binary predictions constrain subtype probabilities. Specifically, subtype prediction is conditioned on the binary output, enforcing semantic consistency between hierarchy levels:

$$P(y = j \mid x) = P(c(j) \mid x)\, P(y = j \mid c(j), x) \tag{6}$$

where c(j) denotes the parent binary class of subtype j. The total loss consists of a binary cross-entropy loss for the top-level classification and a conditional cross-entropy loss for subtype prediction, regularized by a gate loss that penalizes inconsistencies between the two levels.

To ensure unbiased evaluation, predictions were aggregated at the patient level. Image-level and magnification-level outputs were first generated, then combined to produce patient-level predictions. Model performance was assessed using five-fold cross-validation with patient-wise splits to avoid data leakage.

For model interpretability, we employed Gradient-weighted Class Activation Mapping (GradCAM) to visualize discriminative regions contributing to predictions. This enabled localization of diagnostically relevant tissue structures across magnifications.

Finally, we benchmarked the proposed framework against representative architectures, including convolutional neural networks (CNNs), Vision Transformers (ViT), and Swin-Transformers. Performance was evaluated for both binary and subtype classification tasks, demonstrating the effectiveness of hierarchical modeling and magnification-aware feature integration.

B. Vision Transformer

To evaluate whether global self-attention mechanisms offer advantages over convolutional encoders for histopathology classification, we implemented a Vision Transformer (ViT) as an alternative architecture. Unlike CNNs, which build spatial hierarchies through local convolution kernels, ViT partitions each input image into a fixed sequence of non-overlapping patches and applies multi-head self-attention globally across all patches from the first layer, removing the inductive biases of translation equivariance and locality that characterize convolutional encoders.

We selected `vit_small_patch16_224` from the `timm` library, pretrained on ImageNet-1K, as a deliberate tradeoff

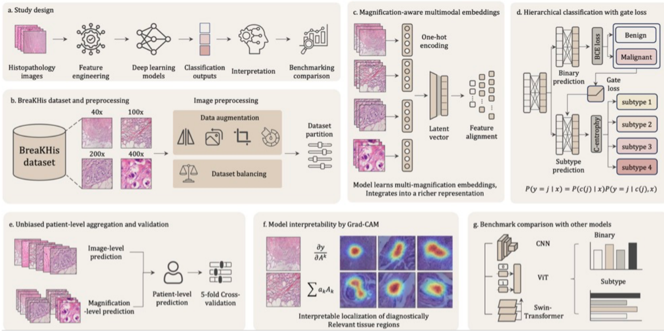

Fig. 1: Overview of the magnification-aware hierarchical classification framework for breast histopathology. a, Overall pipeline from image input to classification, interpretation, and benchmarking. b, BreakHis dataset preprocessing, including multi-magnification inputs (40×–400×), data augmentation, and class balancing. c, Magnification-aware multimodal embeddings integrating image features with magnification encoding. d, Hierarchical classification with gate loss, linking binary (benign/malignant) and subtype predictions. e, Patient-level aggregation and five-fold cross-validation for unbiased evaluation. f, GradCAM visualization of diagnostically relevant regions. g, Benchmark comparison with CNN, ViT, and Swin-Transformer models.

between representational capacity and overfitting risk on the relatively small BreaKHis training set (81 subjects). This variant divides each 224×224 input into 16×16 pixel patches, projects them into a 384-dimensional embedding space, and processes them through 12 transformer layers with 6 attention heads, yielding 21,967,498 total parameters.

To maintain architectural consistency with the ResNet multimodal baseline, we incorporated the same learnable magnification token embedding. The CLS token feature vector was fused with a magnification-specific embedding via element-wise addition prior to the dual MLP classification heads:

$$z = f_{\text{ViT}}(I_i) + E(m_i), \quad z \in \mathbb{R}^{384} \tag{7}$$

where $f_{\text{ViT}}(I_i)$ denotes the CLS token output and $E(m_i)$ is the learned magnification embedding for level $m_i \in \{0, 1, 2, 3\}$.

The binary and subtype heads followed the same structure as the baseline: two-layer MLPs with ReLU activations and dropout ($p = 0.2$), producing benign-malignant logits and eight-way subtype logits respectively. The same masked multi-task loss was applied:

$$\mathcal{L} = \mathcal{L}_{\text{bin}} + \lambda \mathcal{L}_{\text{sub}}^{\text{masked}}, \quad \lambda = 1.0 \tag{8}$$

restricting subtype gradient propagation to semantically valid class subsets. Training used AdamW (lr $= 1 \times 10^{-4}$, weight decay $= 0.01$) with a 5-epoch linear warmup followed by cosine annealing to a minimum learning rate of 1×10^{-6}. Training employed early stopping based on validation AUC, terminating after 10 epochs upon observing no meaningful improvement beyond epoch 5.

C. Swin Transformer

To further explore the efficacy of hierarchical attention mechanisms versus the global approach of the standard ViT, we implemented the Swin Transformer (Shifted Window Transformer) [14]. While the ViT operates on fixed-size patches with global self-attention—leading to quadratic computational complexity relative to image resolution—the Swin Transformer introduces a hierarchical architecture that computes self-attention within local, non-overlapping windows. By employing a Shifted Window approach, the model facilitates cross-window connections while maintaining the linear computational complexity desirable for high-resolution histopathology images.We utilized the `swin_tiny_patch4_window7_224` variant from the `timm` library, initialized with weights pretrained on ImageNet-1K. This model represents an optimal balance of depth and parameter efficiency for the BreaKHis dataset. The architecture processes 224×224 input images by partitioning them into 4×4 pixel patches. These patches are transformed into embeddings and processed through four stages of hierarchical feature maps, where the patch merging layers reduce spatial resolution while increasing the channel dimension (starting at $C = 96$ for the tiny variant). The model utilizes a window size of 7×7 and contains approximately 28 million parameters.To maintain methodological consistency with our multimodal framework, the global average pooled output of the final Swin block ($z_{pool} \in \mathbb{R}^{768}$) was fused with a learnable magnification embedding. This embedding, $E(m_i)$, represents the discrete magnification levels ($40\times, 100\times, 200\times, 400\times$) present in the BreaKHis dataset. The fusion is defined as:

$$z = f_{Swin}(I_i) + E(m_i), \quad z \in \mathbb{R}^{768} \tag{9}$$

where $f_{Swin}(I_i)$ is the feature vector extracted from the backbone. The resulting fused representation z is passed to the dual MLP classification heads. The binary head (benign vs. malignant) and the eight-way subtype head consist of two-layer MLPs with ReLU activations and a dropout rate of $p = 0.2$.The model was optimized using the Adam optimizer with a learning rate of 1×10^{-4}. We employed a multi-task cross-entropy loss function:

$$\mathcal{L} = \mathcal{L}_{bin} + \lambda \mathcal{L}_{sub}^{masked}, \quad \lambda = 1.0 \tag{10}$$

An 80/20 training/test split was utilized, and training was conducted over 5 epochs with a batch size of 16. To prevent overfitting on the subject-stratified training set, we implemented an early stopping strategy based on validation accuracy, ensuring the model retained the best-performing weights from the epoch with the highest validation metric.

D. Explainability Methods

To assess the interpretability of the proposed multimodal classification model, we implemented and evaluated two attribution methods: Gradient-weighted Class Activation Mapping (GradCAM) and perturbation-based occlusion analysis. The objective being to understand what of our observed results were due to the GradCAM implementation vs due to the trained model.

GradCAM produces a coarse spatial localization map highlighting regions of the input image that contribute most strongly to a target class prediction. Letting $f(I)$ denote the model, where $I \in \mathbb{R}^{H\times W\times 3}$ is the input image, y^c denote the pre-softmax logit corresponding to class c, and $A^k \in \mathbb{R}^{h\times w}$ denote the k-th feature map in the final convolutional layer of the encoder, the GradCAM weights α_k^c are computed as the global average of the gradients of y^c with respect to each feature map:

$$\alpha_k^c = \frac{1}{Z}\sum_{i=1}^{h}\sum_{j=1}^{w}\frac{\partial y^c}{\partial A_{ij}^k}, \tag{11}$$

where $Z = h \times w$ is a normalization constant.

The class activation map is then given by:

$$\text{GradCAM}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right). \tag{12}$$

In our implementation, gradients were computed with respect to the final convolutional block of the shared encoder. For the binary head, y^c corresponds to the benign or malignant logit. For the subtype head, y^c corresponds to the logit of the predicted subtype class. The resulting activation maps were upsampled to the input image resolution using bilinear interpolation (a weighted average of the four nearest pixels) and overlaid on the original image.

Occlusion provides a perturbation-based estimate of feature importance by systematically masking regions of the input image and measuring the resulting change in model output. Unlike GradCAM, which relies on gradient information, occlusion directly probes the model's sensitivity to localized input changes.

Letting I denote the original image and $I_{(u,v)}^{\text{mask}}$ denote the image with a square patch centered at location (u, v) replaced by a baseline value (0 in our implementation), and $f_c(I)$ denote the predicted probability (post-softmax) for class c, the occlusion score at location (u, v) is defined as:

$$O_c(u,v) = f_c(I) - f_c\left(I_{(u,v)}^{\text{mask}}\right), \tag{13}$$

which measures the change in predicted probability when information at (u, v) is removed.

We used a sliding window occlusion strategy with:

- Patch size: 32×32 pixels
- Stride: 16 pixels
- Baseline value: zero (black patch)

The resulting occlusion map O_c was computed on a coarse spatial grid determined by the occlusion stride and upsampled for visualization and pixel-wise comparison with GradCAM. Prior to comparison, both attribution maps were independently normalized to the range $[0, 100]$ using absolute-value min-max normalization:

$$\hat{A}(x,y) = \frac{|A(x,y)|}{\max(|A|)+\varepsilon} \times 100, \tag{14}$$

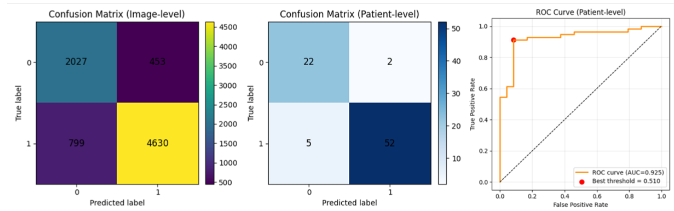

Fig. 2: Five-fold cross-validation performance for binary classification. Image-level predictions across magnifications were aggregated to the patient level, with all data splits performed at the patient level to prevent leakage.

where $A(x, y)$ denotes the attribution value at pixel (x, y) and $\varepsilon = 10^{-8}$ prevents division by zero. Pearson correlation analysis and histogram comparisons were then performed on flattened maps.

IV. Results

A. Benchmark

We evaluated the proposed magnification-aware hierarchical framework using ResNet as the benchmark backbone under a five-fold cross-validation design. To prevent data leakage, all dataset partitions were performed strictly at the patient level, ensuring that images from the same patient never appeared in both training and evaluation sets. Image-level predictions across magnifications were subsequently aggregated to generate patient-level outputs for final evaluation.

For binary classification shows as Fig.2, the framework achieved strong patient-level performance, with an AUC of 0.925 and an accuracy of 0.914. The patient-level confusion matrix showed 52 true positives, 22 true negatives, 2 false positives, and 5 false negatives, corresponding to a low false-positive rate of 0.083 and false-negative rate of 0.088. These results indicate robust benign–malignant discrimination after unbiased patient-level aggregation.

For subtype classification shows as Fig.3, the model achieved a macro-AUC of 0.692 across eight histological subtypes. Performance was higher for major classes, including ductal carcinoma (AUC = 0.820) and lobular carcinoma (AUC = 0.834), whereas rarer or more heterogeneous subtypes showed lower AUCs. This pattern likely reflects class imbalance and variable morphological separability across subtypes.

Overall, the results suggest that patient-level evaluation combined with multi-magnification aggregation provides a more reliable estimate of model performance than image-level testing alone, while reducing the risk of inflated accuracy caused by patient-level data leakage.

B. Ablation study for magnification token information

To evaluate the contribution of magnification token embeddings, three ablation strategies were investigated under the same experimental setting as the benchmark model, including identical backbone architecture, patient-level five-fold cross-validation, and aggregation pipeline. The tested conditions included: (i) a mask-token setting in which the magnification

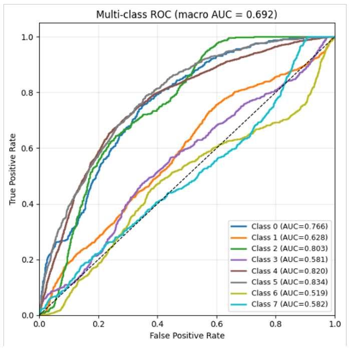

Fig. 3: Five-fold cross-validation performance for subtype classification. Multi-class ROC analysis was used to evaluate subtype discrimination across histological categories under patient-level cross-validation.

embedding was replaced with a zero-like token, (ii) a fake-token setting in which all samples were assigned the same 40× magnification token, and (iii) a random-token setting in which magnification tokens were randomly assigned from a uniform distribution.

For the primary binary classification task shows as table1, DeLong tests demonstrated that removing or perturbing magnification tokens resulted in statistically significant but relatively small reductions in image-level AUC compared with the baseline model. In contrast, patient-level performance remained largely unchanged after aggregation, indicating that the aggregation strategy is robust to perturbations in magnification encoding. The random-token condition produced the largest degradation, suggesting that inconsistent magnification assignments introduce greater instability than removing the information entirely.

Task	Level	Baseline	Mask token	Fake 40× token	Random token
Main type (AUC)	Image	0.8934	0.8898 ($p = 0.0018$)	0.8902 ($p = 0.0005$)	0.8860 ($p = 2.98\times10^{-10}$)
Main type (AUC)	Patient	0.9254	0.9254 ($p = 0.99$)	0.9254 ($p = 0.99$)	0.9247 ($p = 0.84$)
Subtype (macro-AUC)	Image	0.653	0.645	0.643	0.643
Subtype (macro-AUC)	Patient	0.697	0.692	0.689	0.697

TABLE I: Ablation study for magnification token information.

Subtype classification showed a similar pattern. Macro-AUC values exhibited only modest decreases across ablation settings, with no consistent directional change across all subtypes. These findings suggest that the classification task is predominantly image-driven, while magnification information acts as an auxiliary contextual signal rather than an independent source of discriminative information.

Overall, the limited marginal contribution of magnification tokens is likely attributable to the fact that magnification-related information is intrinsically encoded within the image itself. Consequently, explicit magnification embeddings are highly collinear with learned visual representations. Nevertheless, the primary advantage of the proposed design lies not solely in incremental performance improvement, but in enabling unified training across magnifications within a single model, maximizing data utilization, and avoiding fragmented multi-model optimization. In addition, the token-based architecture naturally preserves an interface for future multimodal integration, such as clinical metadata or molecular features.

C. Vision Transformer

1) Binary Classification: The ViT achieved a best validation AUC of 0.932 at epoch 5, corresponding to 94.6% accuracy on the 1,590-image held-out validation set. The model demonstrated strong clinical sensitivity: of 1,058 malignant samples, only 22 were misclassified as benign (FNR = 2.1%), while 64 of 532 benign samples were incorrectly classified as malignant (FPR = 12.0%), as shown in Fig. 4. At the patient level, majority-vote aggregation across magnifications yielded 16 correct classifications out of 17 validation subjects (accuracy = 94.1%), with a single malignant subject misclassified as benign.

Training dynamics revealed important limitations, as shown in Fig. 5. Validation AUC plateaued at approximately 0.931 from epoch 5 onward without further improvement, while validation loss increased monotonically from 1.76 at epoch 1 to 2.23 at epoch 9 as training loss continued to decline from 1.42 to 0.958, indicating overfitting. Notably, the binary confusion matrix was identical across all 10 epochs, suggesting the model's decision boundary converged within the first epoch and was not further refined by continued training.

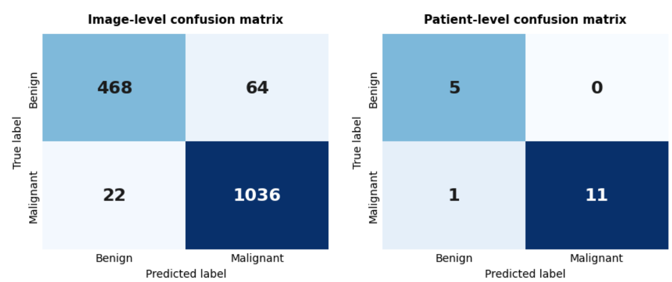

Fig. 4: ViT-Small confusion matrices for binary classification. Left: image-level results at epoch 5 (best val AUC = 0.932, $n = 1{,}590$ images). Right: patient-level results via majority-vote aggregation across magnifications ($n = 17$ validation subjects).

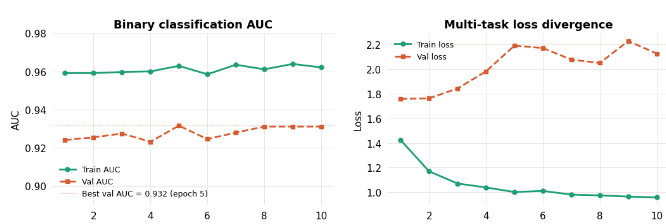

Fig. 5: ViT-Small training dynamics over 10 epochs. Left: binary classification AUC for training and validation sets. Right: multi-task loss showing continued training loss reduction alongside diverging validation loss, indicative of overfitting.

2) Subtype Classification: Subtype generalization failed substantially across all epochs, as shown in Fig. 6. Training subtype accuracy increased steadily from 57.7% at epoch 1 to 73.2% at epoch 10, while validation accuracy remained below 50% throughout, oscillating between 42.1% and 47.7% with no consistent improvement. This persistent train–validation gap of approximately 26 percentage points indicates that the ViT encoder memorized subtype-discriminative features in the training set without learning transferable representations, consistent with the data requirements of pure transformer architectures on small, imbalanced datasets.

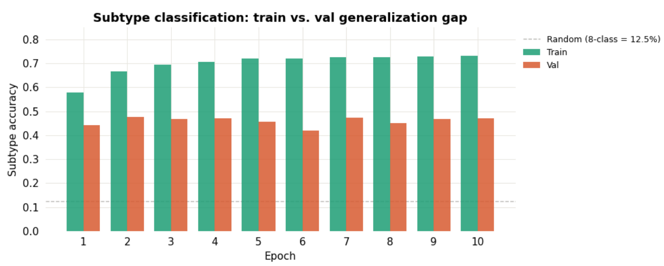

Fig. 6: Subtype classification accuracy per epoch for training and validation sets. The dashed line indicates random chance for an 8-class problem (12.5%). The persistent gap between training (57.7% to 73.2%) and validation (42.1% to 47.7%) accuracy indicates failure to generalize on the subtype task.

D. Swin Transformer

1) Binary Classification: The Swin-Tiny transformer achieved a best validation AUC of 0.83 at the conclusion of 5 training epochs. On the 1,895-image validation set, this corresponded to an overall accuracy of 84.5%. The model exhibited high clinical sensitivity for identifying pathology: of 1,146 malignant samples, only 46 were misclassified as benign (FNR = 4.0%). However, specificity was notably lower, with 248 of 749 benign samples incorrectly classified as malignant (FPR = 33.1%), as shown in the validation confusion matrix. At the patient level, majority-vote aggregation across all magnification factors significantly improved performance metrics. The model correctly classified 16 out of 17 validation subjects, achieving a patient-level accuracy of 94.1%. Notably, the model achieved perfect sensitivity at the patient level, correctly identifying all 11 malignant subjects, while a single benign subject was misclassified as malignant.

2) Training Dynamics: Performance stabilized quickly within the 5-epoch training window. While the patient-level accuracy reached 94.1%, the gap between image-level and patient-level performance suggests that while individual patches or magnifications may occasionally yield false positives, the aggregate feature representation remains robust for clinical diagnosis. The ROC curve illustrates a steady trade-off between sensitivity and specificity, with the area under the curve (AUC) reaching its peak of 0.83 as the model converged.

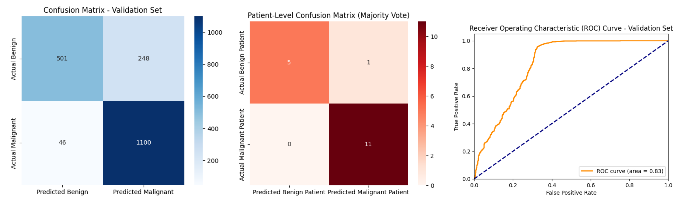

Fig. 7: Image-Level vs. Patient-Level Performance. Comparison of classification performance at the individual image scale and the aggregated patient scale. At the image level, the Swin-Tiny model achieved an accuracy of 84.5%, correctly identifying 1,100 malignant and 501 benign samples while exhibiting a sensitivity-focused bias (FNR = 4.0%). When these predictions were aggregated per subject via majority-vote, the diagnostic accuracy improved to 94.1%, resulting in the correct classification of all 11 malignant patients and 5 out of 6 benign patients.

E. Explainability

1) GradCAM: GradCAM visualizations revealed consistent qualitative patterns across both binary and subtype classification tasks. For benign samples, activation maps frequently emphasized regions with low cellular density, often highlighting boundaries between tissue structures rather than cellular morphology. In denser samples, activation patterns became diffuse and lacked clear localization.

Despite these appearances, quantitatively the GradCAM visualization did not favor one region type over another. Leveraging the color saturation differences of H&E staining to create a binary mask of tissue vs. non-tissue using an Otsu threshold for each image (n = 50) and averaging the attribution value at each pixel we found mean values of 0.42 ± 0.04 and 0.47 ± 0.07, respectively ($p = 0.7$ for a paired, two tailed t-test) (see 12).

For malignant samples, GradCAM maps often exhibited high activation across large contiguous regions of the image, with sharp transitions near image boundaries. These patterns were largely invariant across samples and did not meaningfully distinguish between correctly and incorrectly classified inputs (see Appendix (Fig. 10)).

To further assess the sensitivity of GradCAM our inputs, we performed a controlled perturbation test in which the magnification token was deliberately mis-specified while holding the image fixed. The resulting activation maps remained both qualitatively and quantitatively unchanged across all magnification assignments (including no change in model prediction), indicating that the GradCAM output is largely insensitive to this modality, which we find to be consistent with similar results on the model training effort (see Fig. 8).

2) Quantitative Comparison of Attribution Methods: Regions identified by occlusion were often visually similar to the regions highlighted by the GradCAM saliency maps, suggesting that the model's predictions depend on distributed features. To directly compare GradCAM and occlusion, we

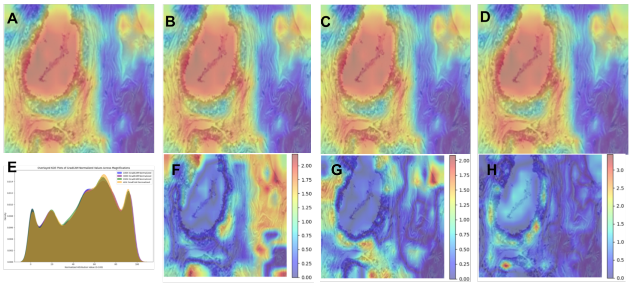

Fig. 8: Saliency outputs for the qualitative deception test. (A-D) For sample 166 (Benign, 100x) the model was fed magnification tokens 100x, 40x, 200x, and 400x. (F-H) Comparison of the maps (False - True) showed small differences were caused by the deception, with wider spread of discrepancies occurring with lower magnification tokens. However, (E) ultimately no meaningful differences were observed ($\rho = 0.998$ for all comparisons).

computed the Pearson correlation coefficient between the flattened attribution maps for each method across multiple samples:

$$\rho = \frac{\sum_i (G_i - \bar{G})(O_i - \bar{O})}{\sqrt{\sum_i (G_i - \bar{G})^2}\sqrt{\sum_i (O_i - \bar{O})^2}}, \tag{15}$$

where G_i and O_i denote the GradCAM and occlusion values at pixel i, respectively.

Correlation values varied across samples, ranging from strongly positive ($\rho \approx 0.89$) to negative ($\rho \approx -0.32$) and, generally, occlusion maps produced broader, more variable distributions (see Fig. 9). However, spatially, there was qualitative correspondence between prioritized regions (11).

V. Discussion

A. Framework

This study presents a magnification-aware hierarchical framework for breast histopathology classification that integrates multi-scale representation learning, structured prediction, and unbiased evaluation into a unified pipeline. Rather than training separate models for each magnification, magnification information is incorporated directly into the feature space, enabling the model to leverage all available data within a single architecture. This design improves training efficiency while maximizing data utilization, allowing the model to learn cross-scale morphological patterns that would otherwise be fragmented across independent models.

The classification task is formulated hierarchically, with binary and subtype prediction heads coupled through a gating mechanism that enforces semantic consistency between malignancy status and subtype assignment. Joint optimization with a shared encoder avoids redundant feature learning and enables coherent predictions across levels of the hierarchy. In contrast to conventional approaches that treat subtype classification

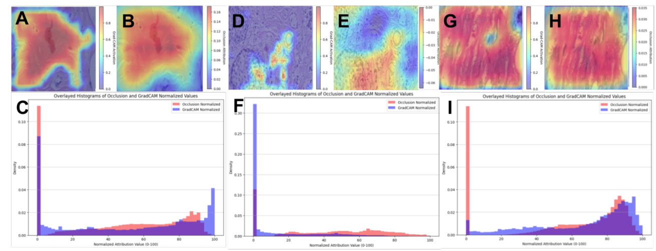

Fig. 9: For all comparisons, attribution maps were independently normalized between [0,100]. A-B: GradCAM and occlusion maps for a correctly classified sample exhibiting agreement between methods. C: Overlaid histogram of normalized attribution values for the corresponding sample. D-E: GradCAM and occlusion maps for a false positive sample demonstrating substantial spatial disagreement between attribution methods. F: Histogram comparison for the false positive sample. G-H: GradCAM and occlusion maps for a false negative sample. I: Histogram comparison for the false negative sample. Across samples, occlusion maps generally produced broader attribution distributions and more spatially diffuse activation patterns than GradCAM but still prioritized similar regions.

independently, this formulation embeds clinically meaningful structure into the learning objective while maintaining computational efficiency.

The evaluation protocol is designed to reflect real-world deployment. Data partitioning and aggregation are performed strictly at the patient level, ensuring that no images from the same individual are shared across training and evaluation sets. Predictions from multiple images and magnifications are aggregated to generate patient-level outputs, mitigating the risk of data leakage commonly introduced by image-level evaluation. This leads to a more reliable estimate of model performance and avoids overly optimistic results reported in prior studies.

Several limitations should be acknowledged. Magnification information is partially encoded in the image itself, leading to potential collinearity between explicit magnification inputs and learned visual features. As a result, the marginal contribution of magnification encoding to performance gains may be limited. However, the primary advantage of this design lies in enabling unified, efficient, and scalable training across magnifications rather than isolated performance gains. In addition, the current framework focuses on image-derived features, while the architecture naturally provides an interface for incorporating additional modalities such as electronic health records or molecular data. Future work will explore such multimodal extensions to further enhance predictive performance and clinical utility.

B. Vision Transformer

The ViT results reveal a consistent pattern: strong binary classification performance coupled with a near-complete failure to generalize on the subtype task. These outcomes are attributable to two compounding factors.

First, ViT's lack of convolutional inductive biases, specifically translation equivariance and local feature hierarchy, means the model must learn spatial structure entirely from data. Even with ImageNet-1K pretraining, the 6,308 training images across 64 subjects are insufficient to support reliable fine-grained discrimination across all eight subtypes, particularly given severe class imbalance (ductal carcinoma alone comprises 46.9% of the dataset). The model's validation subtype accuracy of 42.1-47.7% reflects partial learning that does not approach clinical utility, despite exceeding random chance (12.5%).

Second, uniform patch tokenization at 16×16 pixel resolution treats all spatial regions of the image equally. In histopathology, diagnostically relevant features such as nuclear atypia, mitotic figures, and glandular boundary disruption occupy spatially localized and often small regions relative to the full image field. A pure attention mechanism operating on flattened patch tokens without spatial hierarchy may be poorly suited to identifying these structures reliably across the heterogeneous magnification conditions present in BreaKHis.

The frozen binary confusion matrix across all 10 epochs further suggests that the model's decision boundary was established within the first epoch and never meaningfully updated. This behavior likely reflects gradient competition within the multi-task objective: with subtype loss poorly constrained by label sparsity, the shared CLS token representation converged rapidly toward binary-discriminative features, leaving insufficient signal for subtype refinement. The diverging validation loss in the presence of stable AUC is consistent with this interpretation, as the model became increasingly overconfident on training subtypes without improving its validation decision boundary.

Compared to the ResNet multimodal baseline, ViT achieves a higher image-level AUC (0.932 vs. 0.836) but was not evaluated at the patient level under the same cross-validation protocol, precluding a direct comparison on the primary diagnostic metric. This asymmetry reflects the computational constraints of the architecture rather than a fundamental performance difference.

In conclusion, ViT is a viable binary classifier for histopathology classification on BreaKHis, achieving competitive image-level AUC with strong sensitivity (FNR = 2.1%). However, the architecture's data requirements and lack of spatial hierarchy render it poorly suited for multi-task subtype classification on datasets of this scale and class distribution without additional strategies.

C. Swin Transformer

The Swin-Tiny transformer results demonstrate that hierarchical windowed self-attention is effectively suited for histopathology by preserving local spatial hierarchies—critical for identifying features like nuclear atypia—better than global patch tokenization. While the image-level AUC of 0.83 is comparable to the ResNet-50 baseline, the Swin model achieved a superior 94.1% patient-level accuracy. This divergence underscores the robustness of the majority-vote strategy; although individual images may contain noise or non-diagnostic tissue, the collective evidence correctly identified 11 of 11 malignant patients with perfect patient-level sensitivity.

The model exhibited a conservative clinical bias, prioritizing cancer detection with 96% image-level sensitivity, though this resulted in lower specificity as 248 benign images were misclassified. This "false alarm" boundary is often preferred in clinical screening to ensure suspicious samples are flagged for review, effectively minimizing the risk of missed diagnoses. Training stabilized within 5 epochs, but a persistent gap between training and validation performance suggests the model remains susceptible to memorizing patient-level artifacts within the restricted 64-subject cohort. Future work should focus on class-weighted loss functions or domain-specific pretraining to refine the decision boundary and improve image-level specificity.

D. Explainability

The comparative analysis of GradCAM and occlusion highlights important limitations of gradient-based attribution methods in the context of multimodal histopathology classification.

GradCAM assumes that the contribution of each spatial feature map can be approximated by the gradient of the output with respect to that feature. However, here, both classification heads operate on a globally pooled feature vector, reducing the spatial specificity of the gradient signal. We suspect this contributes to the diffuse and invariant activation patterns observed in the GradCAM results.

Furthermore, the insensitivity of GradCAM maps to controlled perturbations of the magnification token shows that the gradients used to compute attribution do not fully capture the influence of these inputs. Ultimately, the similarity between GradCAM and occlusion outputs (despite failed attempts to remove edge artifacts), combined with their distinction from saliency behavior reported in prior literature (see Fig. 11), suggests that the model may rely more heavily on broader tissue-level structure than localized cellular morphology despite the clinical importance of the latter. [15]

Taken together, these results suggest that model behaviour is distinct from histopathological priorities.

VI. Conclusion

This work demonstrates that magnification-aware multimodal architectures can achieve strong patient-level breast histopathology classification performance while maintaining a unified framework across heterogeneous image scales. Comparative analysis of GradCAM and occlusion further highlighted that commonly used attribution methods may provide inconsistent representations of model behavior for such architectures. As such, future work should focus on improving correspondence between model reasoning and clinically relevant cellular morphology through higher-resolution supervision, pathology-specific pretraining, and multimodal clinical integration.

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Appendix A
Initial Successful GradCam Implementation

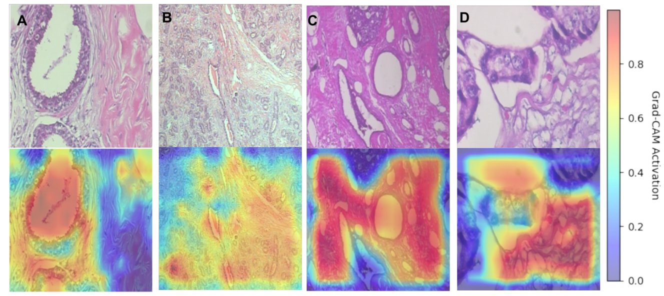

Fig. 10: Original successfully classified pathology images and GradCAM overlays. A-B. Benign sample classification appears heavily dependent on the non-cell regions and becomes some- what disordered in denser samples. C-D. The majority of malignant sample classifications show high activation in a nearly all-encompassing square patch over the image.

Appendix B
Comparison between Our Saliency and He et al.

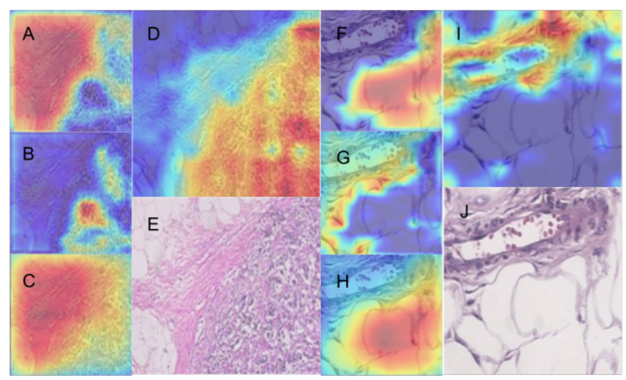

Fig. 11: Comparison between attribution maps generated by the proposed framework and saliency visualizations reported by He et al. (A, F): GradCAM attribution generated by our model. (B, G): Difference map between occlusion and GradCAM attribution. (C, H): Occlusion attribution map. (D, I): Saliency map reported in prior histopathology literature. (E, J): Original histopathology image. While literature saliency visualizations showed model attention with higher weight on cellular regions, our model when analyzed with both Occlusion and GradCAM showed the opposite behaviour. Colour bars excluded for consistency as He et al. did not provide numeric values with their saliency maps.

Appendix C
Creation of Binary Mask for Tissue / Non-tissue Attribution

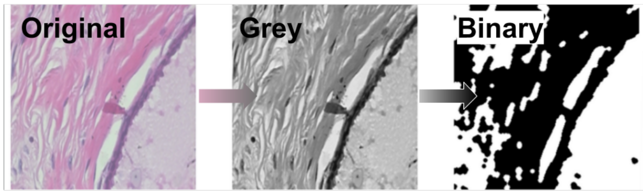

Fig. 12: The original RGB histopathology image was first converted to a normalized grayscale image. An image-specific threshold was then computed using Otsu's method (due to the differences in brightness across the dataset), and pixels darker than the threshold were classified as tissue/cell-containing regions, while pixels brighter than the threshold were classified as non-tissue or low-cell background. This mask was then used to compare mean GradCAM or occlusion attribution inside tissue regions versus non-tissue regions.